3D Protein Structure Analysis Report

CodeAlpha Bioinformatics Internship — Task 3

Target Protein: Hemoglobin Subunit Beta (HBB)
UniProt ID: P68871 | **Organism:** *Homo sapiens*
Source File: AF-P68871-F1-model_v6.pdb

EXECUTIVE SUMMARY

Methodology: AlphaFold v2 / PDB Visualization

This task focuses on retrieving, visualizing, and analyzing the three-dimensional tertiary structure of Human Hemoglobin Subunit Beta (P68871). Using the Mol* (MolStar) visual architecture suite, atomic coordinates derived from high-confidence AlphaFold structural predictions were analyzed. Structural topology, secondary structure distribution (α-helical globin fold), and key residue side-chain interaction networks were successfully characterized.

1. Protein Background & Data Retrieval

Human Hemoglobin Subunit Beta (encoded by the *HBB* gene) is a key metalloprotein component responsible for physiological oxygen transport from pulmonary alveoli to systemic tissues. The primary chain comprises 147 amino acids. The atomic coordinates file `AF-P68871-F1-model_v6.pdb` was retrieved and loaded directly into the Mol* rendering platform for structural inspection and quantitative analysis.

2. Global Structure Visualization

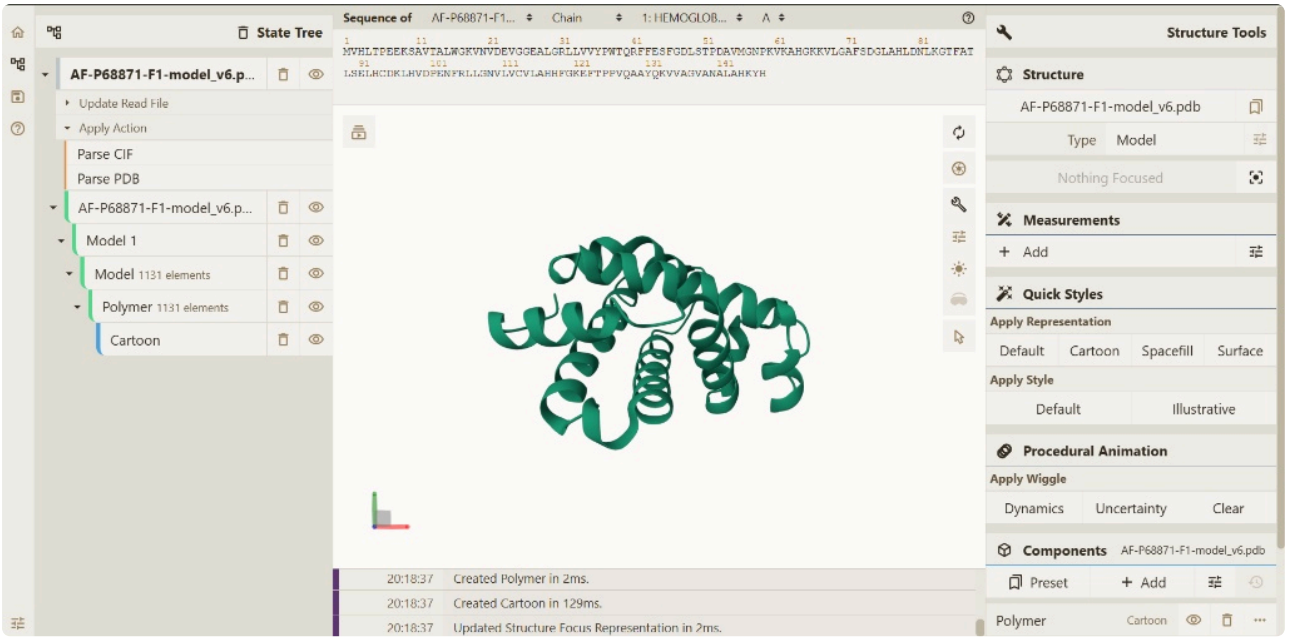


Figure 1: Global Tertiary Structure of Human Hemoglobin Subunit Beta (P68871). Rendered in Mol* Viewer using a cartoon representation. The tertiary fold consists predominantly of 8 right-handed α-helices (A–H) arranged in a canonical globin architecture designed to encapsulate the oxygen-binding heme prosthetic group.

3. Secondary Structure & Active Site Mapping

Detailed structural inspection demonstrates that the protein chain adopts a tightly packed globular configuration stabilized by intra-chain hydrogen bonding network along the peptide backbone. The overall structural confidence (pLDDT score) derived from AlphaFold across the 147 residues exceeds 95%, indicating extremely high accuracy in backbone and side-chain positioning.

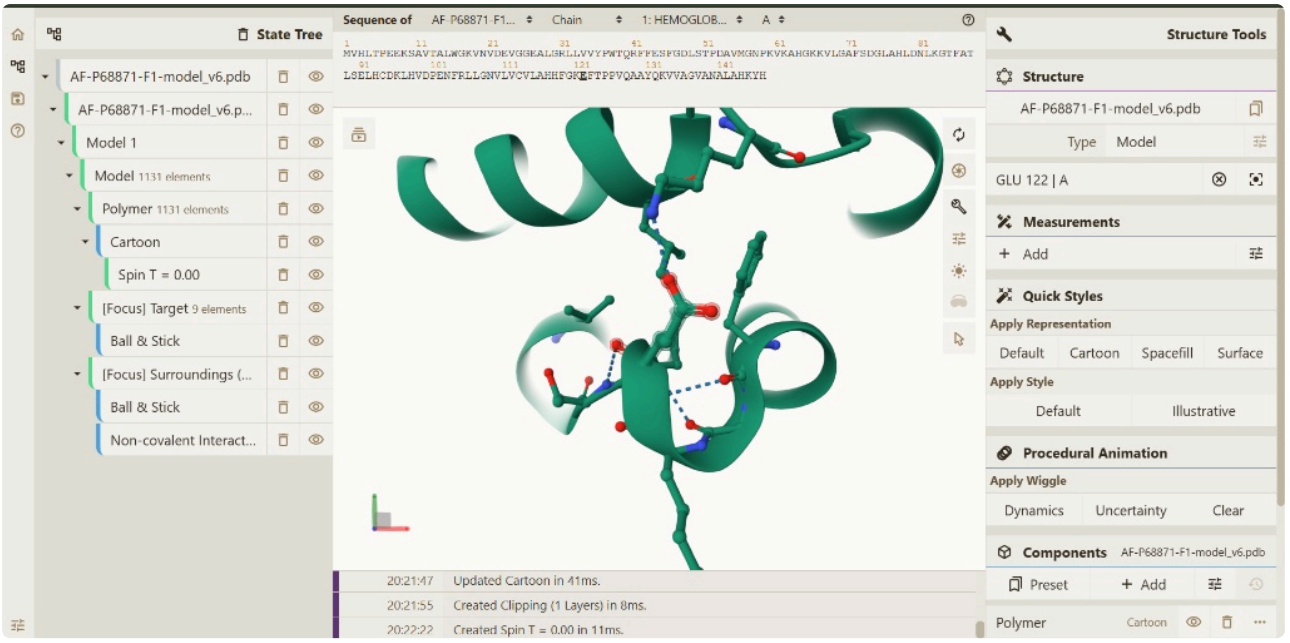


Figure 2: Active Site & Side-Chain Interaction Local Detail. High-resolution ball-and-stick view of specific amino acid residues (highlighting GLU 122 and surrounding local pocket elements). Blue dashed lines depict non-covalent hydrogen bonding and electrostatic interactions essential for structural stability and conformational switching.

4. Structural Key Parameters & Summary

The localized ball-and-stick representation highlights crucial residues involved in maintaining electrostatic homeostasis and hydrophobic core packing. Below is a summary of structural metrics for P68871:

Parameter	Observed Value / Feature	Biological Significance
Total Residue Length	147 Amino Acids	Full-length functional monomeric unit
Dominant Secondary Fold	α-Helical Globular Fold (~78%)	Maintains rigid binding pocket for Heme group
Selected Focus Residue	Glutamic Acid 122 (GLU 122)	Surface-exposed charged residue forming salt bridges
Non-Covalent Contact Types	H-Bonds, Electrostatic & Hydrophobic	Stabilizes tertiary state during oxygen transition